

The Abiotic Methane Ceiling Sets a Floor on Detectable Biospheres

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ABSTRACT

Flux based biosignature assessment compares an inferred surface methane flux against a ceiling on abiotic production. The ceiling in current use is a deliberately conservative upper bound, adopted as a single number. It stands above every other published maximum for the same process, by factors of four to fifty-five, above anything its own generating equation yields at modern Earth crustal production rates, and a factor of three below Earth’s own biological methane flux. The choice of ceiling therefore fixes the smallest biosphere an observation can reveal, and the span of defensible choices is wider than the spectroscopic uncertainty of the retrieval it is compared against.

1. A BOUND DOING THE WORK OF A LIKELIHOOD

N. F. Wogan et al. (2026) infer surface gas fluxes for an Archean-like TRAPPIST-1e by inverting a coupled photochemical and climate model, then ask whether the inferred fluxes are consistent with life. The step from flux to biosignature runs through a set definition: methane fluxes above 3.7×10^{10} molecules $\text{cm}^{-2} \text{s}^{-1}$ are taken to exceed what ocean bottom serpentinization can supply. That value is the 99.7th percentile of the distribution of maximum abiotic fluxes computed by J. Krissansen-Totton et al. (2018), adopted, as N. F. Wogan et al. (2026) say, for simplicity.

As a set boundary the choice is sound and conservative by design. Its consequences appear at the following step, where the complement of the biosignature region is taken as the abiotic alternative and the resulting odds ratio is used as a proxy for the Bayes factor between life and no life. A bound chosen to be generous then does the work of a likelihood.

The construction carries a second consequence. The oxygen biosignature is defined as the subset of $\mathcal{H}_{\text{CH}_4}$ with a net positive oxygen flux, and the life-consistent region $\mathcal{L}$ as its union with the methane biosignature, which is $\mathcal{H}_{\text{CH}_4}$ with the reducing volcanic false positive $\mathcal{V}$ removed. The union therefore restores part of $\mathcal{V}$: their Table 2 masses give $P(\mathcal{V} \cap \mathcal{B}_{\text{O}_2}) = P(\mathcal{L}) - P(\mathcal{H}_{\text{CH}_4}) + P(\mathcal{V})$, at least half of $\mathcal{V}$ in both prior and posterior once two-decimal rounding is propagated. What one branch excludes as a false positive, the other partly readmits.

2. LOCATING THE CEILING

Three comparisons place it, and the first needs no view on what Earth actually produces, since it sets the ceiling against other published maxima. A. Guzmán-Marmolejo et al. (2013) modelled methane production by serpentinization from the kinetics of the reactions involved, with carbon dioxide as the limiting reactant, and report a maximum surface flux of 6.8×10^8 molecules $\text{cm}^{-2} \text{s}^{-1}$ for an Earth mass planet, 1.74 dex below the ceiling. A. S. Merdith et al. (2020) give an upper estimate in which the methane flux approaches the hydrogen flux, 1.12 dex below it. Even the Archean maximum of J. Fiebig et al. (2007), which describes a more vigorous Earth than the one being modelled, falls 0.60 dex short. No published maximum for this process reaches the adopted ceiling, and the three fall short of it by factors of four, thirteen and fifty-five.

The second uses the generating equation itself. The lower corner of the four parameter ranges sampled by J. Krissansen-Totton et al. (2018) is built entirely from Earth calibrated values: their modern crustal production rate, 10 wt% FeO in newly produced crust, the 0.25% conversion of FeO to hydrogen that they calibrate against observation, and the hydrogen to methane ratio of 12 measured in ultramafic hosted vent fluids. Their Equation 6 then returns $0.012 \text{ Tmol yr}^{-1}$, or $\log F_{\text{CH}_4} = 7.67$, a little below the directly measured mid-ocean ridge values. Their distribution runs from what Earth does to the most generous case they were willing to entertain, and the adopted percentile sits 2.90 dex above its lower end.

The third asks what that equation reaches without leaving Earth-like conditions. Holding crustal production at the modern Earth value and setting the other three factors to the generous end of their published ranges, Equation 6 gives

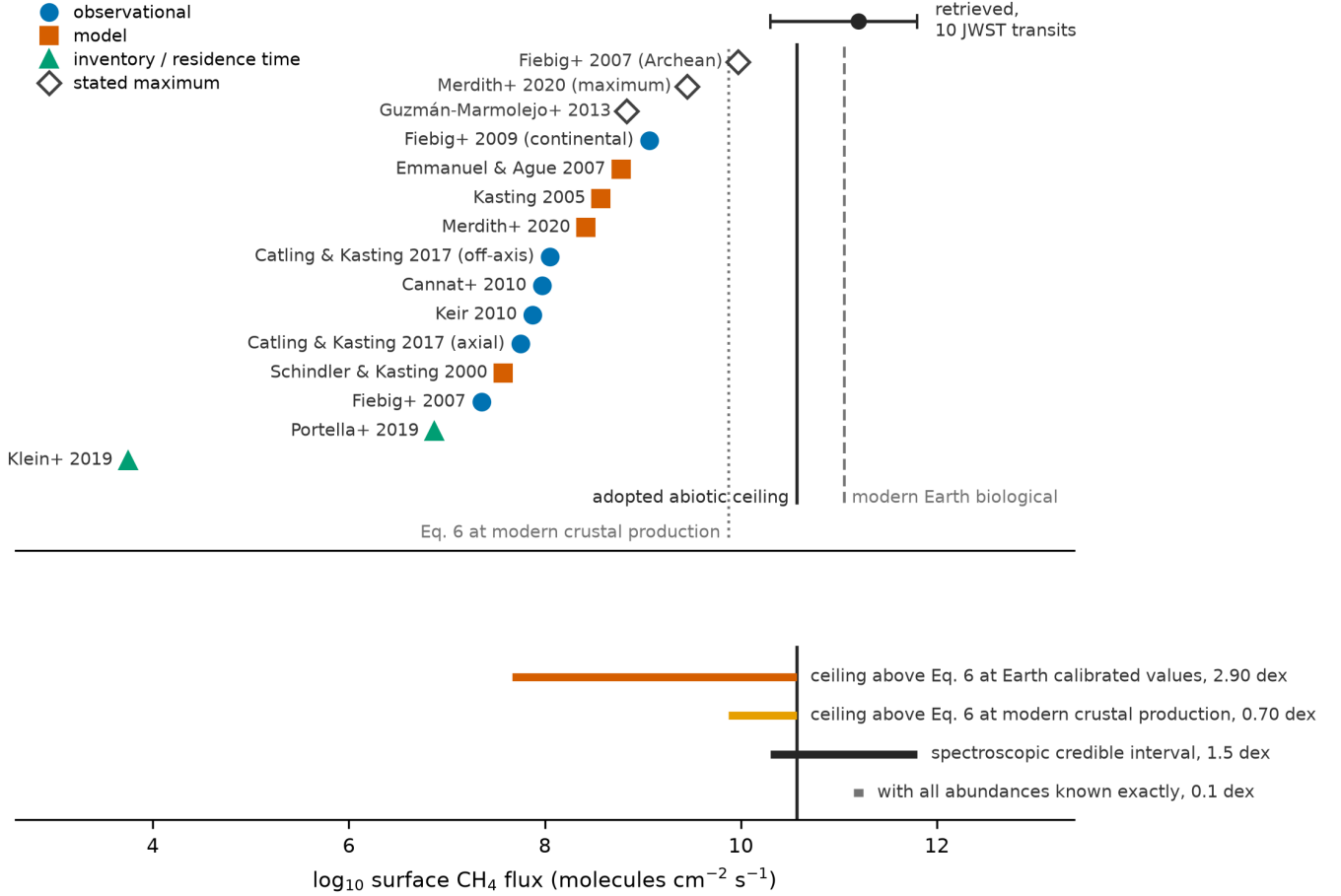


Figure 1. Published estimates of Earth’s abiotic methane production, against the ceiling adopted in flux based biosignature assessment. Filled markers distinguish observational estimates, model estimates, and estimates obtained by dividing a reservoir inventory by a residence time. Open diamonds are the published maxima for this process (J. Fiebig et al. 2007; A. Guzmán-Marmolejo et al. 2013; A. S. Meredith et al. 2020); the ceiling lies above all of them. The solid line is the adopted abiotic ceiling, the dotted line is what Equation 6 of J. Krissansen-Totton et al. (2018) reaches at modern Earth crustal production with its other factors maximal, and the dashed line is modern Earth’s biological methane flux. The point with error bars is the flux retrieved from a synthetic ten transit JWST spectrum of an Archean-like TRAPPIST-1e (N. F. Wogan et al. 2026). The lower panel sets two framings of the ceiling ambiguity against the credible interval of that retrieval and against the narrowest interval attainable if all gas abundances were known exactly.

$\log F_{\text{CH}_4} = 9.87$. The adopted ceiling is 0.70 dex above that, so it is reachable only by invoking crustal production above modern Earth’s. The factor of ten allowed in their range is an early Earth provision, and J. Krissansen-Totton et al. (2018) note that whether such rates are likely for terrestrial exoplanets is an open question.

Figure 1 sets all of this against published estimates of Earth’s realised abiotic methane production (T. L. Schindler & J. F. Kasting 2000; J. F. Kasting 2005; S. Emmanuel & J. J. Ague 2007; J. Fiebig et al. 2007, 2009; M. Cannat et al. 2010; R. S. Keir 2010; D. C. Catling & J. F. Kasting 2017; F. Klein et al. 2019; Y. de Melo Portella et al. 2019; A. S. Meredith et al. 2020), compiled from the supplementary material of M. A. Thompson et al. (2022) and from A. S. Meredith et al. (2020). Every estimate for modern Earth lies below the ceiling, the largest, from continental hydrothermal emissions, by 1.5 dex. Six of the nine seafloor estimates, spanning mid-ocean ridges, off-axis vent fields, submarine volcanism and subduction related sites, cluster between $\log F_{\text{CH}_4} = 7.3$ and 8.1; the remaining three lie between 8.4 and 8.8.

3. WHAT THE CEILING COSTS

The first cost is a floor on detectability. N. F. Wogan et al. (2026) adopt 1.122×10^{11} molecules $\text{cm}^{-2} \text{s}^{-1}$ for modern Earth’s biological methane flux, following D. C. Catling & J. F. Kasting (2017). The ceiling lies a factor of three below that. A planet whose biosphere produced methane at less than a third of Earth’s rate could not be called inhabited on this criterion at any signal to noise. Placing the ceiling at the Earth calibrated value instead moves that floor to roughly one part in 2400 of Earth’s biological flux.

The second cost is that this choice, rather than the spectrum, sets the scale of the uncertainty. The retrieval returns a 68% credible interval 1.5 dex wide for $\log F_{\text{CH}_4}$. Table A1 of N. F. Wogan et al. (2026) gives that width under every combination of gas abundances fixed to their true values, and the narrowest entry is 0.1 dex, the limit of what perfect abundance knowledge would buy on this target. The comparisons in Section 2 span 0.60 to 2.90 dex. The two largest exceed the interval the retrieval actually achieved, the rest are a third to three quarters of it, and every one of them is six to thirty times the 0.1 dex floor. A posterior width and a range of candidate thresholds are different objects, but both enter the same decision in the same units, and on this target they are of comparable size.

4. REPORTING THE DEPENDENCE

Reading the quoted credible interval endpoints as the 16th and 84th percentiles and 11.2 as the median fixes a split normal. It returns 0.76 for the posterior mass above the adopted ceiling, against the 0.77 that N. F. Wogan et al. (2026) tabulate. Across the ceilings considered here that mass runs from 0.65, with Equation 6 at all four factors maximal, through 0.76 at the adopted percentile, to above 0.9 at any published maximum. Only the first two fall inside the quoted interval; the others are extrapolations and give scale rather than exact values.

The biosignature probability can therefore be reported as a function of the assumed ceiling, from a posterior that already exists. The Bayes factor cannot: that needs the prior mass at each candidate ceiling as well. N. F. Wogan et al. (2026) point the same way, noting that the inhabited and lifeless flux regions should more realistically overlap and writing the generalisation in which the sharp set becomes a weighting function $p(\text{life} | F)$.

5. LIMITATIONS

The estimates in Figure 1 describe different reservoirs that are only partly additive, and several of the seafloor values are alternative estimates of one quantity rather than separate contributions to it, so the figure should be read as a range of published values and not as a distribution. They also differ in what they measure: the mid-ocean ridge values of R. S. Keir (2010) and M. Cannat et al. (2010) are fluxes released at vents, whereas A. S. Merdith et al. (2020) model production from the iron oxidation state of serpentinite and note that much of it is consumed in the subsurface, which is why their values sit higher. A ceiling for a planet other than Earth would also need its crustal production rate scaled, as A. Guzmán-Marmolejo et al. (2013) do for a five Earth mass planet.

6. SOFTWARE AND DATA

The compiled estimates, the arithmetic reproducing every value quoted above, and the script that draws Figure 1 are available at <https://github.com/ssiddhantsharma/abiotic-ch4>.

REFERENCES

- | | |
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| Cannat, M., Fontaine, F., & Escartin, J. 2010, in <i>Diversity of Hydrothermal Systems on Slow Spreading Ocean Ridges</i> , ed. P. A. Rona, C. W. Devey, J. Dymant, & B. J. Murton, Vol. 188 (American Geophysical Union), 241–264 | Emmanuel, S., & Ague, J. J. 2007, <i>Geophysical Research Letters</i> , 34, L15810, doi: 10.1029/2007GL030532 |
| Catling, D. C., & Kasting, J. F. 2017, <i>Atmospheric Evolution on Inhabited and Lifeless Worlds</i> (Cambridge University Press) | Fiebig, J., Woodland, A. B., D’Alessandro, W., & Püttmann, W. 2009, <i>Geology</i> , 37, 495 |
| de Melo Portella, Y., Zaccarini, F., & Etiope, G. 2019, <i>Minerals</i> , 9 | Fiebig, J., Woodland, A. B., Spangenberg, J., & Oschmann, W. 2007, <i>Geochimica et Cosmochimica Acta</i> , 71, 3028 |
| | Guzmán-Marmolejo, A., Segura, A., & Escobar-Briones, E. 2013, <i>Astrobiology</i> , 13, 550, doi: 10.1089/ast.2012.0817 |
| | Kasting, J. F. 2005, <i>Precambrian Research</i> , 137, 119 |
| | Keir, R. S. 2010, <i>Geophysical Research Letters</i> , 37, L24609 |

- Klein, F., Grozeva, N. G., & Seewald, J. S. 2019, Proceedings of the National Academy of Sciences, 116, 17666
- Krissansen-Totton, J., Olson, S., & Catling, D. C. 2018, Science Advances, 4, eaao5747, doi: [10.1126/sciadv.aao5747](https://doi.org/10.1126/sciadv.aao5747)
- Merdith, A. S., del Real, P. G., Daniel, I., et al. 2020, Geochemistry, Geophysics, Geosystems, 21, e2019GC008869, doi: [10.1029/2019GC008869](https://doi.org/10.1029/2019GC008869)
- Schindler, T. L., & Kasting, J. F. 2000, Icarus, 145, 262
- Thompson, M. A., Krissansen-Totton, J., Wogan, N., Telus, M., & Fortney, J. J. 2022, Proceedings of the National Academy of Sciences, 119, e2117933119, doi: [10.1073/pnas.2117933119](https://doi.org/10.1073/pnas.2117933119)
- Wogan, N. F., Batalha, N. E., Krissansen-Totton, J., et al. 2026, ApJ, doi: [10.3847/1538-4357/ae6646](https://doi.org/10.3847/1538-4357/ae6646)